

Q1) let =  
P = It is raining  
Q = It is cold

Then:

$$P \wedge Q$$

Q2) let:  
P = the server is down  
Q = website is unreachable

Then:

$$P \rightarrow Q$$

Q3) let:  
P = the file is corrupted  
Q = the password is wrong

Then

$$P \vee Q$$

Q4) let:  
P = The light is on  
Q = the switch is up

Then  $P \rightarrow Q$

Q5] let:  
P = the switch is up  
Q = the light is on

Then:  
 $P \rightarrow Q$

Q6] let  
P = you pass this course  
Q = you submit the project

Then:  
 $P \leftrightarrow Q$

Q7] let  
P = the database  
Q = The cache failed

Then:  
 $\neg (P \wedge Q)$

Q8] let  
P = Neither the router  
Q = Nor the modem is faulty

then  $\neg P \vee \neg Q$

Then:  $P \rightarrow (Q \vee R)$

Q9] let  
P = the test fails  
Q = the code has a bug  
R = the environment is misconfigured.

Q 10]

let

$P$  = we will go hiking

$Q$  = it rains

Then

$$\neg Q \rightarrow P$$



Scanned with CamScanner

Q11)  $P = \text{Alarm is set}$   
 $Q = \text{door locked}$   
 $R = \text{window closed}$   
 $(P \rightarrow Q) \wedge (Q \rightarrow R)$

Q12)  $P = \text{Sam attends}$   
 $Q = \text{Pet attends}$   
 $R = \text{Quin attends}$   
 $(P \rightarrow Q) \wedge (P \rightarrow \neg Q)$

Q13)  $P = \text{Printer 1 online}$   
 $Q = \text{printer 2 online}$   
 $(P \vee Q) \wedge \neg (P \wedge Q)$

- 14)  $P$  : Backup 1 is succeeded  
 $Q$  : Backup 2 is succeeded  
 $R_n$  : Backup n succeeded

$$\text{~~the~~ } P \vee Q \vee \dots \vee R_n$$

- 15)  $P$  = Logging is verbose  
 $Q$  = feature flag is enabled  
 $P \leftrightarrow Q$

- 16)  $P$  = Code reviewed  
 $Q$  = It complies  
 $R$  = Dependencies installed  
 $(P \rightarrow Q) \wedge (Q \rightarrow R)$

- 17)  $P$  : meeting today  
 $Q$  : Alex available  
 $R$  : Blake available  
 $\text{~~the~~ } P \rightarrow (\neg Q \wedge \neg R) \vee P \rightarrow \neg (Q \vee R)$

- 18)  $P$  = sensor triggers  
 $Q$  = Battery charged  
 $R$  = Siren sounds  
 $(P \wedge Q) \rightarrow R$



19)  $P =$  License expires  
 $Q =$  software start  
 $R =$  Trial mode is active  
$$P \rightarrow (\neg Q \vee R) (\vee P \rightarrow (Q \rightarrow R))$$

20)  $P \div$  Data is transmitted  
 $Q =$  Network secure  
 $P \rightarrow Q$